Status	Finished
Started	Wednesday, 21 October 2026, 11:28 AM
Completed	Wednesday, 21 October 2026, 11:47 AM
Duration	19 mins 16 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search can be applied to both sorted and unsorted lists. ✓
- b. Linear search is more efficient than binary search on large lists.
- c. Linear search divides the list into halves.
- d. Linear search requires the list to be sorted.

Question **2**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It can be applied only if the list is sorted ✓
- b. It compares elements sequentially
- c. It works on unsorted lists
- d. It always starts from the beginning of the list

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. There must be a mechanism to access middle element directly
- b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- d. Must use a sorted array

Question **4**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search starts over from the beginning
c. The search continues until the end of the list
d. The search stops immediately ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓
b. The last element is returned
c. An error is raised
d. The first element is returned

Question **6**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n-1
c. log n
d. n ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It can only be applied to large lists
c. It requires the list to be sorted ✓
d. It does not work with negative numbers

Question **8**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the search algorithm finishes
c. When the element is found in the list ✓
d. When the list is sorted

Question **9**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎
b. O(n log n)
c. O(log n)
d. O(1) ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓
b. log n
c. n/2
d. 1

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By sorting the list first
c. By dividing the list into halves
d. By comparing each element in the list sequentially ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Deleting elements from a list
- c. Inserting elements into a list
- d. Sorting elements in a list

Question **13**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the left sublist
- b. At the beginning of the list
- c. In the right sublist ✓
- d. In the middle of the list

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By starting from the first element
- b. By comparing each element sequentially
- c. By dividing the list length by two ✓
- d. By using a hash function

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. Number values should only be present

  Number values should only be present

- b. The list must be sorted
- c. There should be direct access to the middle element in any sublist
- d. There must be a mechanism to delete and/or insert elements in the list ✓